

A Preliminary Study for Developing Perceived ESG Scale to Measure Public Perception Toward Organizations' ESG Performance

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Presenter **Dayeoun Jang**

What is ESG?



Environment



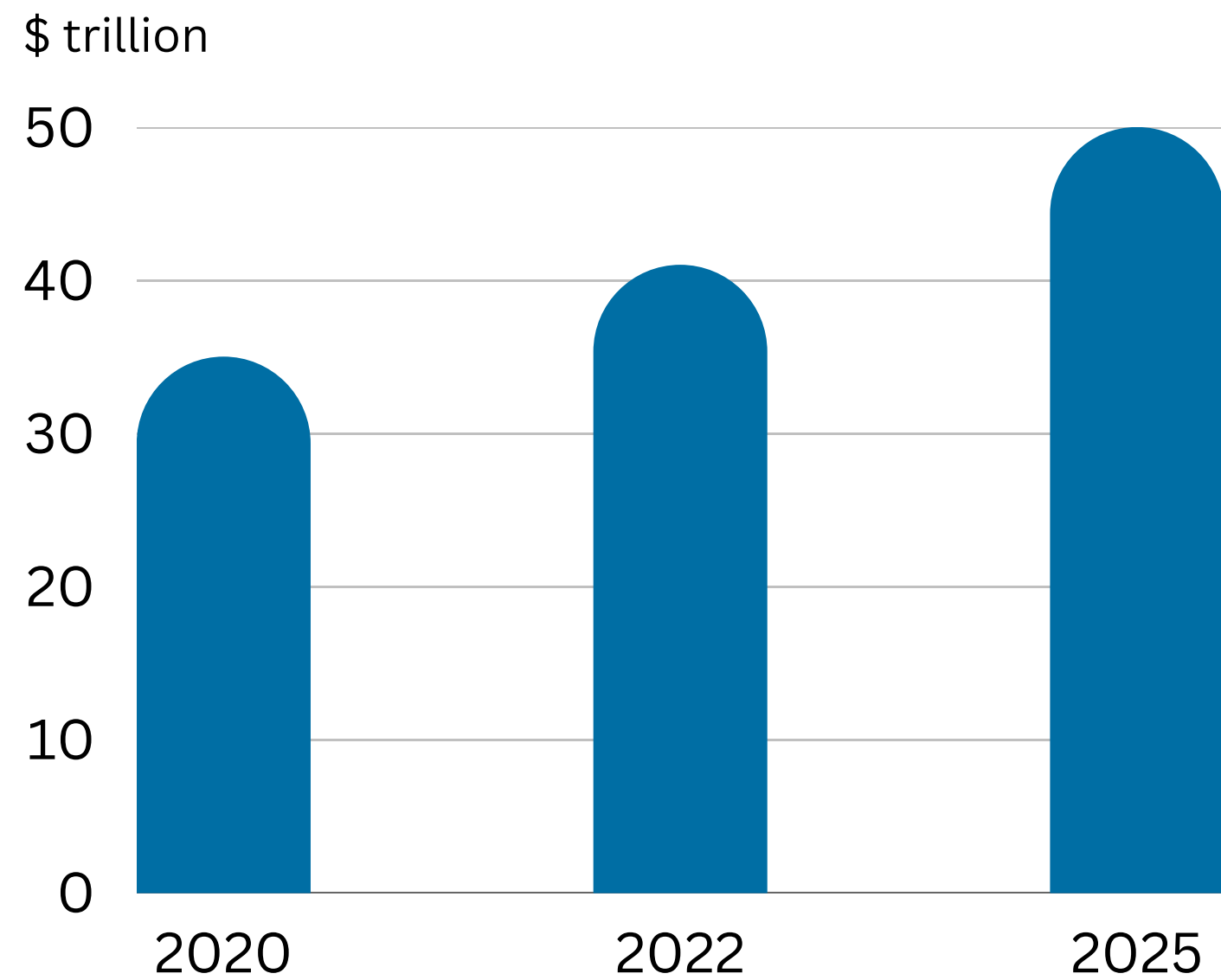
Society



Governance

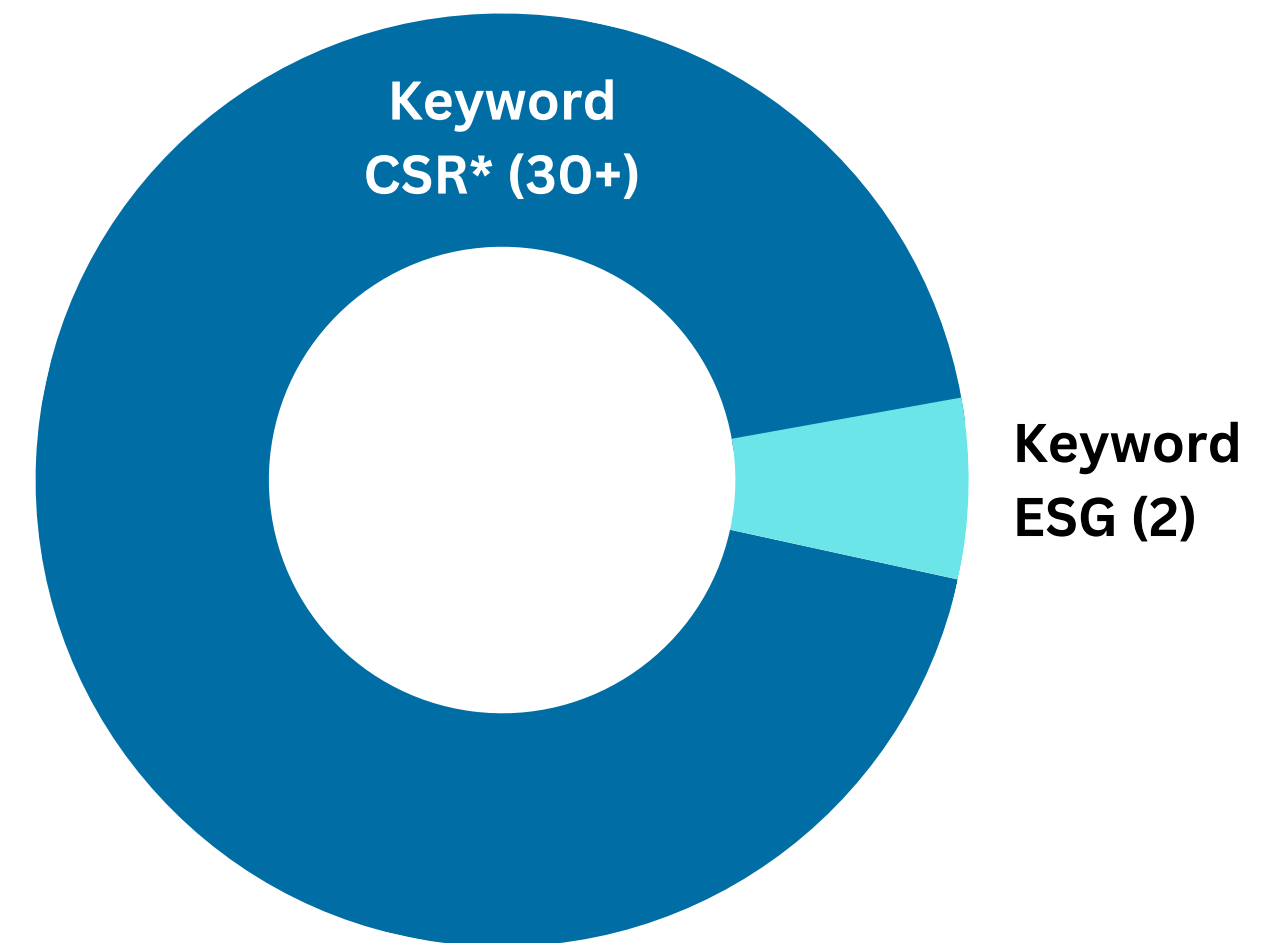
Non-financial factors that can affect an organization's reputation and long-term success

Global ESG assets



Source: Bloomberg (2022)

Number of studies in Public Relations Reviews (2022)



*Corporate social responsibility

CSR vs ESG

ESG includes broad aspects of CSR principles with organizational activities associated with governance and sustainability

ESG suggests intuitive standards for companies and the public by naming three clear categories

ESG in PR RESEARCH

A unique research opportunity for scholars

The relationship between media coverage of an organization's ESG and its financial performance (Kölbel et al., 2017; Naumer & Yurtoglu, 2020)

ESG news coverage affects an organization's performance more than non-ESG news coverage.

ESG in PR PRACTICE

Public perception is crucial in understanding an organization's ESG performance



Public perception toward ESG can impact business outcomes
(Luo & Bhattacharya, 2006; Tetrault Sirsly & Lvina, 2019)



Actual ESG activities $\neq$ Public perception toward ESG
(Du et al., 2010; Eisenegger & Schranz, 2011; Viererbl & Koch, 2022; Yoon et al., 2006)

CURRENT PROBLEM

More ESG study is required in PR research, but...



Current available ESG indices do not focus on public perception



There is no valid and reliable scale in measuring public perception on ESG

METHODS & RESULTS

We followed the procedure for scale development widely suggested and implemented in earlier works (i.e., Churchill Jr, 1979; Hinkin, 1995; Worthington & Whittaker, 2006)

All analyses were carried out using R version 4.2.0

STEP 01

Develop an operational definition of P (perceived)–ESG**Perceived E**

Perception of an organization's eco-friendly management, including management to reduce carbon emissions and engage in eco-friendly production and distribution

**Perceived S**

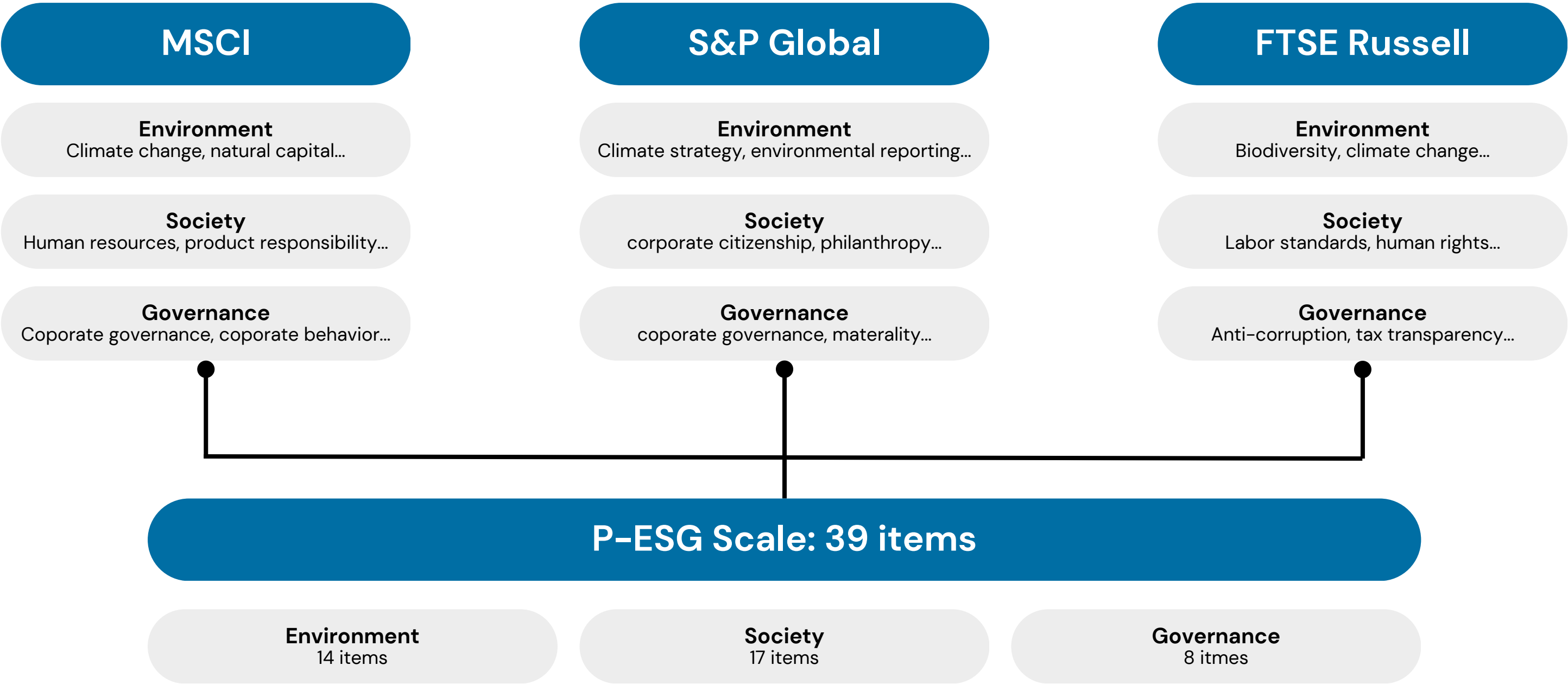
Perception of an organization's socially responsible business conduct, including its guarantee of basic rights in the workplace and its community contribution activities

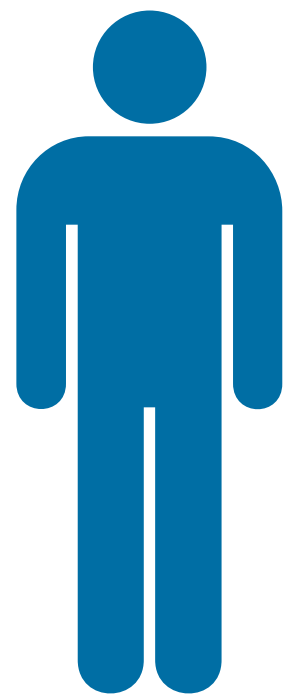
**Perceived G**

Perception of an organization's governance, including its ownership structure and disclosure of information.

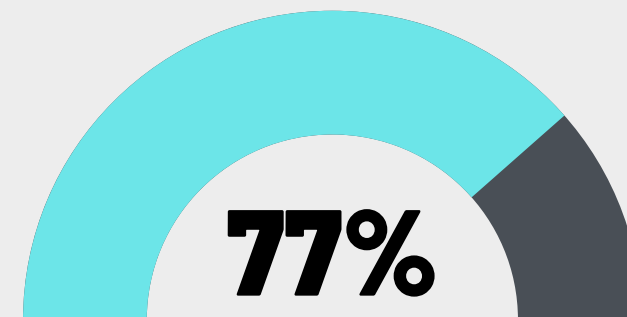
STEP 02

Review the currently available ESG indices



STEP 03**Assess face validity****13 Experts**

Majoring in advertising, public relations,
media informatics, and communication

Ph.D. Students**Average age****35.46**

5 items were dropped

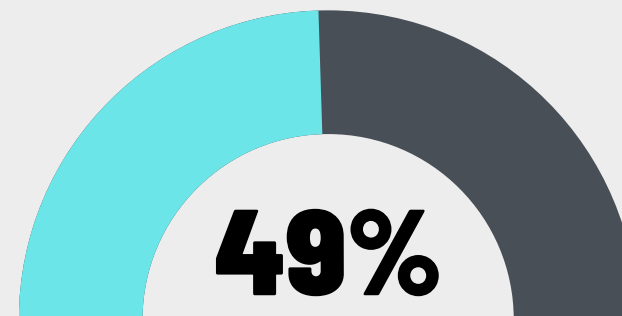
P-ESG Scale: 34 items

STEP 04**Public Survey**

1,644
Participants

Recruited from an online research panel
provided by a Korean research company

Female



Average age

45.4

SD = 11.02

STEP 04

Public Survey



**60
Companies**

- 01 Identified 11 industry sectors using the Global Industry Classification Standard
- 02 Identified 100 leading Korean companies based on stock value
- 03 Selected 60 companies based on the proportion of each industry sector

STEP 04

Public Survey



Measurements

Five-point
Likert scale

01

P-ESG scale

02

CSR scale (Kim et al., 2005)

 $M = 3.37, SD = 0.65$

03

Intention to purchase

 $M = 3.62, SD = 0.86$

04

Intention to invest

 $M = 3.36, SD = 0.91$

The respondents were asked to respond to surveys for two randomly selected Korean companies

Total Collected Data Point: 3,288

STEP 05

Analysis: EFA

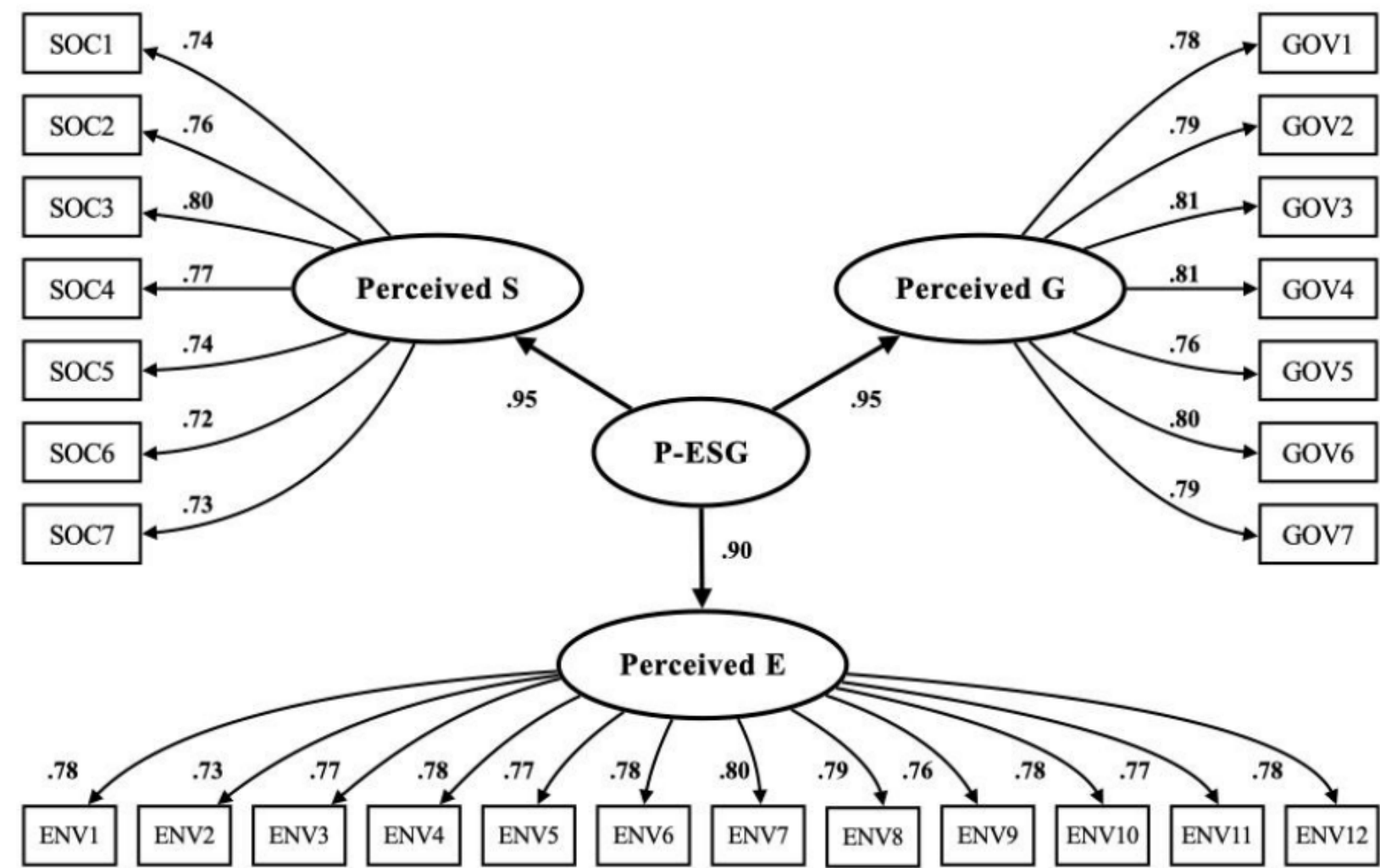
Using principal axis factoring with Promax rotation to check the dimensionality
We dropped 8 items having a factor loading lower than .3

	Extracted Factors		
	Perceived E (12 items)	Perceived S (7 items)	Perceived G (7 itmes)
Eigen value	6.99	4.06	4.65
Proportion of variance	.27	.16	.18
α [95% CI]	.95 [.94, .95]	.90 [.90, .91]	.92 [.92, .93]
ω [95% CI]	.94 [.94, .95]	.90 [.89, .90]	.92 [.91, .92]

STEP 05

Analysis: CFA

A second-order CFA using the maximum likelihood estimation method with bootstrapping



$\chi^2 (296) = 1822.523, p < .01, GFI = .957, AGFI = .949, NFI = .971, CFI = .975, RMSEA = .040, SRMR = .021$

	AVE	CR
E	.602	.948
S	.567	.922
G	.629	.901

STEP 05

Analysis: CFA

A second-order CFA using the maximum likelihood estimation method with bootstrapping

The heterotrait-monotrait ratio (HTMT) of correlations

	E	S	G
E	1		
S	.853	1	
G	.853	.902	1

The upper limits of the 95% CI of the estimated factor correlations

	E	S	G
E	1		
S	.864	1	
G	.862	.910	1

STEP 05

Analysis: Pearson's correlation

Pearson’s correlation analyses were performed for the P-ESG and CSR Scale to examine the concurrent validity of the construct.

Correlations between the P-ESG scale and CSR scale

	M	SD	E	S	G	P-ESG	CSR
E	3.23	0.70	1				
S	3.41	0.66	.79 [.78, .80]	1			
G	3.27	0.70	.80 [.78, .81]	.82 [.81, .83]	1		
P-ESG	3.30	0.64	.93 [.92, .93]	.93 [.93, .94]	.94 [.93, .94]	1	
CSR	3.37	0.65	.80 [.79, .81]	.81 [.80, .82]	.83 [.82, .84]	.87 [.87, .88]	1

STEP 05

Analysis: Regression

A weighted least square (WLS) regression and support vector regression (SVR) were conducted to examine the predictive validity of the P-ESG

Regression analyses of the P-ESG scale and CSR scale for behavioral intentions

		Purchase intention		Investment intention	
		WLS Regression	SVR	WLS Regression	SVR
P-ESG	R ²	.38	.36	.38	.34
	RMSE	0.68	0.70	0.71	0.75
	MAE	0.53	0.51	0.55	0.55
CSR	R ²	.36	.33	.34	.32
	RMSE	0.69	0.72	0.74	0.76
	MAE	0.54	0.53	0.58	0.56

DISCUSSION

Our results confirm that the P-ESG scale is statistically reliable and valid in measuring the public perception of an organization's ESG efforts and performance.

The conceptualization and operationalization of ESG offer a way to measure the public's response to an organization's ESG efforts

By developing a scale to measure the public perception of ESG performance, we suggest that ESG could be an important research topic in PR

The P-ESG scale is well suited for research in contexts other than PR

The findings from this study shed light on the potential role of ESG in reputation management and crisis management

This study proposes a rigorous approach to evaluating the predictive validity of a new scale in PR research

From a practical standpoint, the scale developed in this study could help PR practitioners in gauging their organizations' ESG performance

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Thank you for listening

Presenter **Dayeoun Jang**

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Appendix

Final items of the P-ESG scale		
Scale	<i>M</i>	<i>SD</i>
Environment	3.22	0.71
The company establishes and implements policies to reduce carbon emissions.	3.29	0.82
The company reduces the use of fossil energy and expands the use of renewable energy.	3.19	0.90
The company increases eco-friendly investment and fosters relevant human resources.	3.30	0.91
The company establishes policies to prevent the depletion of natural resources.	3.18	0.88
The company has established regulations to protect our ecosystems.	3.12	0.88
The company produces and sells eco-friendly products and services.	3.24	0.91
The company uses environmentally friendly materials and a sustainable supply chain.	3.25	0.87
The company operates with an eco-friendly production system.	3.28	0.88
The company develops and uses eco-friendly packaging materials.	3.18	0.91
The company strives to reduce industrial waste and expand recycling.	3.20	0.88
The company engages in various actions and social campaigns to protect the environment.	3.31	0.87
The company fulfills eco-friendly practices by voluntarily participating in global environmental programs.	3.27	0.86

Appendix

Final items of the P-ESG scale		
Scale	<i>M</i>	<i>SD</i>
Society	3.41	0.66
The company ensures stable employment and a positive working environment.	3.42	0.85
The company has a healthy corporate culture and positive employee relationships.	3.32	0.85
The company provides a safe and healthy workplace for its employees.	3.33	0.84
The company provides education/support programs for its employees.	3.37	0.81
The company provides good products and services and constantly strives to improve its product quality.	3.49	0.84
The company provides products and services that are safe.	3.52	0.80
The company is well-equipped with consumer remedy systems.	3.38	0.81
Governance	3.27	0.70
The company's ownership/governance structure is stable and transparent.	3.28	0.87
The CEO of the company provides healthy leadership.	3.27	0.89
The board of directors operates transparently and independently of the company's ownership.	3.26	0.85
The company has a transparent system of internal audits.	3.25	0.86
The company protects the rights of shareholders and consistently communicates with them.	3.29	0.82
The employees comply with workplace ethics and anti-corruption guidelines.	3.23	0.85
The company sincerely discloses its financial, business, and ESG performances.	3.31	0.82